PROJECT REPORT

CSA2001: Fundamentals Of AI & ML
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Introduction

The rapid growth of Artificial Intelligence (AI) and Machine Learning (ML) has enabled the development of intelligent systems that can predict outcomes based on data. In the education sector, student performance prediction is an important application of machine learning.

This project focuses on developing a **Student Exam Score Predictor** that estimates a student's exam score based on various factors such as study hours, attendance, mental health, sleep patterns, and part-time job status.

The system helps in understanding how different lifestyle and academic factors influence student performance.

Problem Statement

Many students are unaware of how their daily habits affect their academic performance.

Factors such as:

- Low study time
- Poor attendance
- Lack of sleep
- Mental stress
- Part-time jobs

can negatively impact exam scores.

This project aims to build a machine learning system that predicts exam scores and helps students make better decisions.

Objectives

- To develop a machine learning model for predicting exam scores
- To analyze factors affecting student performance
- To create a CLI-based prediction system
- To build a user-friendly Streamlit web app
- To provide quick and accurate predictions

Methodology

The project follows these steps:

1. **Data Collection**
   Dataset contains student-related features like study hours, attendance, etc.
2. **Data Preprocessing**

    - Convert categorical values (Yes/No → 1/0)
    - Normalize inputs
3. **Model Training**
    - Train model using supervised learning
    - Save model using `.pkl` file
4. **Prediction System**
    - CLI version using Python
    - Web UI using Streamlit

Machine Learning Model

The model used is a **Regression Model** (since output is numeric score).

- Input → Features
- Output → Predicted exam score (0–100)

The model learns relationships between:

- Study hours
- Attendance
- Mental health
- Sleep
- Work status

System Architecture

The system architecture of the Student Exam Score Predictor is designed to process user inputs and generate accurate predictions using a trained machine learning model. The system accepts input parameters such as study hours, attendance, mental health, sleep duration, and part-time job status.

These inputs undergo preprocessing, where categorical data is converted into numerical format and structured into a format suitable for the model. The processed data is then fed into the trained model (`best_model.pkl`), which predicts the exam score.

Finally, the predicted score is displayed to the user through an interactive interface, ensuring ease of use and quick results.

```
┌──────────────────────────────────────────────────────────┐
│      USER INPUT (Study, Attendance, Mental Health,       │
```

```
│                    Sleep, Job)                          │
└─────────────────────────┬──────────────────────────────┘
                                          ▼
┌─────────────────────────┴──────────────────────────────┐
│                  DATA PREPROCESSING                     │
│        (Converting Inputs to Numerical Format)          │
└─────────────────────────┬──────────────────────────────┘
                                          ▼
┌─────────────────────────┴──────────────────────────────┐
│                 TRAINED ML MODEL                        │
│              (Model: best_model.pkl)                    │
└─────────────────────────┬──────────────────────────────┘
                                          ▼
┌─────────────────────────┴──────────────────────────────┐
│               PREDICTED EXAM SCORE                      │
│            (Score Output: 0-100 Range)                  │
└────────────────────────────────────────────────────────┘
                                          ▼
┌─────────────────────────┴──────────────────────────────┐
│                   DISPLAY OUTPUT                        │
│                 (CLI / Web UI App)                      │
└────────────────────────────────────────────────────────┘
```

Figure 1: System Architecture of the Student Exam Score Predictor System

Implementation

The project is implemented using the Python programming language.

The machine learning model is trained using a Jupyter Notebook and saved as a `.pkl` file (`best_model.pkl`). This trained model is then used to make predictions based on user input.

The CLI version of the system is implemented using `main.py`, which takes input from the terminal such as study hours, attendance, mental health, sleep hours, and part-time job status, and provides the predicted exam score.

Additionally, a Streamlit-based user interface (`app.py`) is developed to provide an interactive and user-friendly experience. The web application uses sliders and dropdown menus to collect user input and displays the predicted exam score instantly.

Results

The system successfully predicts student exam scores based on user input. The predictions are displayed in both CLI and web UI formats.

The results demonstrate that the model can effectively analyze academic and lifestyle factors such as study hours, attendance, mental health, sleep, and part-time job status to estimate exam performance. As shown in Figures 2 and 3, the system provides predictions through both command-line and Streamlit-based interfaces, ensuring flexibility and ease of use.

The model provides accurate and consistent predictions based on input parameters, demonstrating the effectiveness of machine learning in student performance analysis.

```
Student Exam Score Predictor
Study Hours: 5
Attendance Percentage: 100
Mental Health (1-10): 6
Sleep Hours: 8
Part-Time Job (Yes/No): No

 Predicted Exam Score: 89.93
```

Figure 2: Command Line Interface Output of Exam Score Prediction

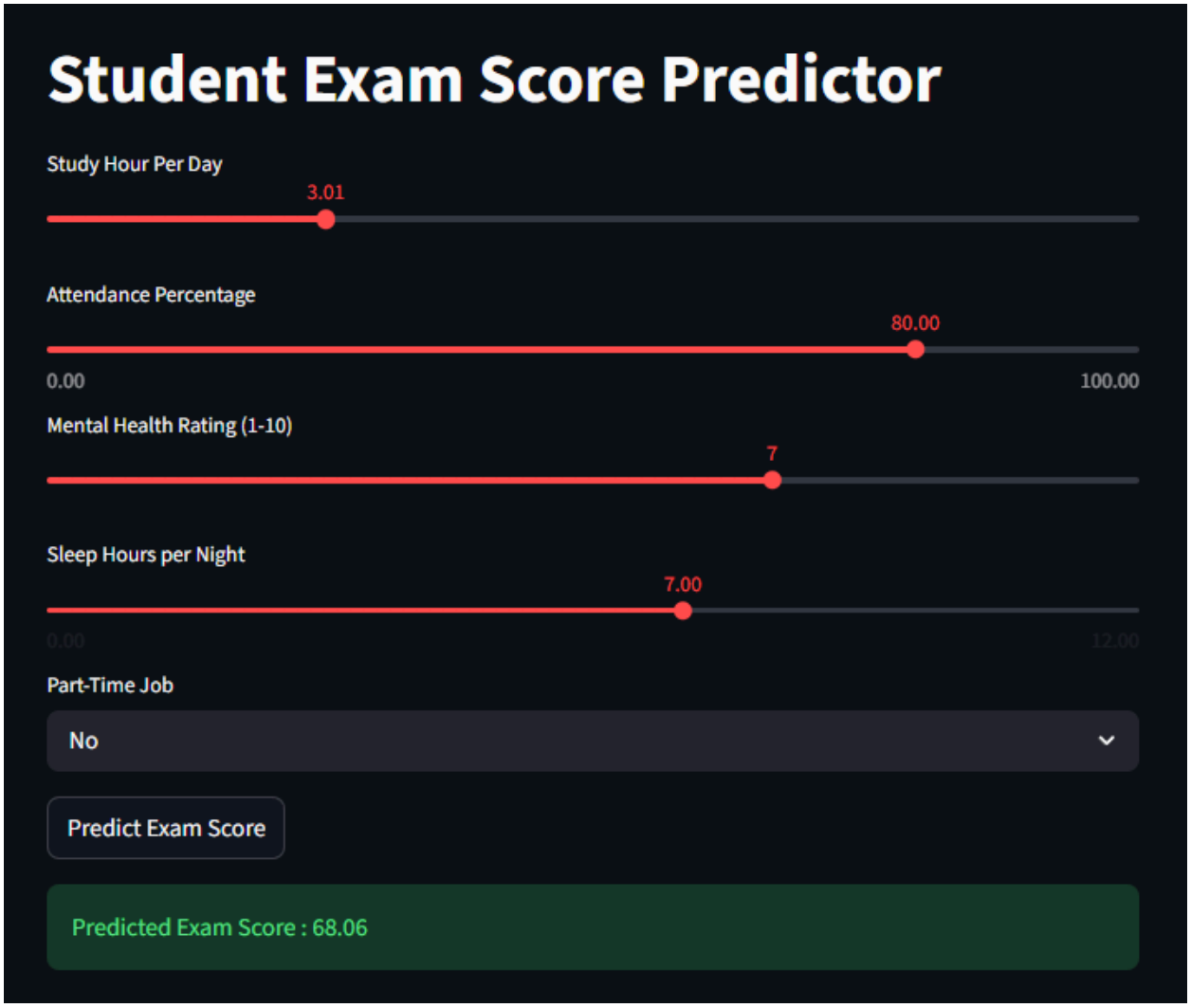

Figure 3: Streamlit-Based User Interface for Exam Score Prediction

Advantages

- Easy to use
- Fast prediction
- Helps students understand performance factors
- Uses machine learning concepts
- Provides user-friendly interface

Limitations

- Uses dataset from Kaggle
- Not 100% accurate
- Limited features
- Depends on input accuracy

Future Scope

- Use real educational datasets
- Improve model accuracy
- Add more student-related parameters (study methods, distractions, etc.)
- Deploy as web/mobile application

Conclusion

This project demonstrates the use of machine learning in predicting student exam scores based on academic and lifestyle factors. It shows how AI can be used to analyze user input and provide useful insights about student performance.

The system helps students understand how factors like study hours, attendance, mental health, and sleep affect their results, encouraging better study habits and decision-making.

This project also fulfills the requirement of CLI-based execution as specified in the course guidelines, along with providing an interactive Streamlit-based user interface.